Chapter 2 ~ Digital Image Fundamentals

Created By
S. M. Mahedy Hasan
Assistant Professor, Dept. of CSE
RUET, Rajshahi-6204, Bangladesh.

Modified by
Md. Manirujjaman
Lecturer, Dept. of CSE
DIU, Satarkul, Badda, Dhaka

Sampling and Quantization

- To create a **digital image**, we need to convert the continuous sensed data into digital form. This involves two processes:
  - ✓ Sampling
  - ✓ Quantization
- An image may be **continuous** with respect to the **x- and y-coordinates**, and also in **amplitude**.
- Digitizing the **coordinate** values is called **sampling**.
- Digitizing the **amplitude** values is called **quantization**.

Sampling and Quantization

- ❖ The one-dimensional function in Fig. (2) is a plot of amplitude (intensity level) values of the continuous image along the line segment AB in Fig. (1).

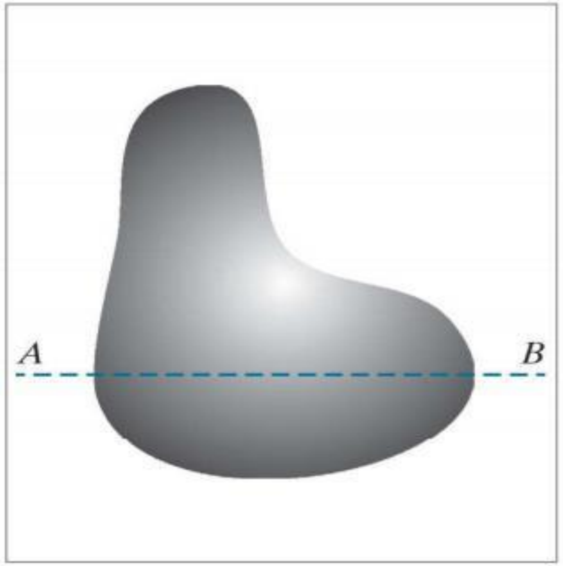


Figure 1: Continuous image

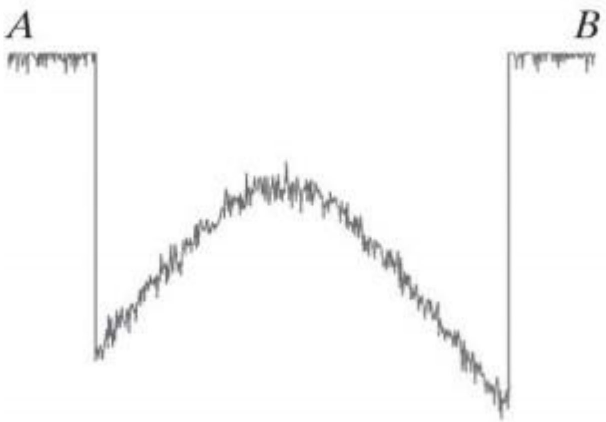


Figure 2: A scan line showing intensity variations along line *AB* in the continuous image.

Sampling and Quantization

- To sample this function, we take equally spaced samples along line AB, as shown in Figure.
- In order to form a digital function, the intensity values also must be converted (quantized) into discrete quantities.
- The right side of Fig. a shows the intensity scale divided into eight discrete intervals, ranging from black to white.
- The digital samples resulting from both sampling and quantization are shown in the following figure.

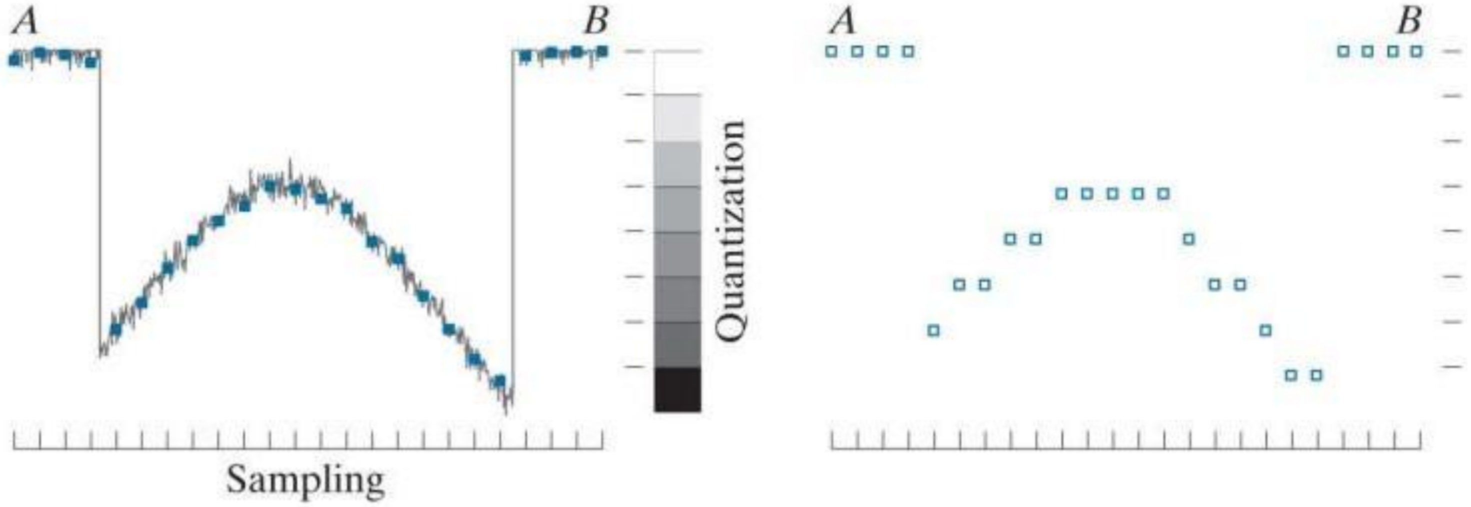


Figure 3: Sampling and Quantization

Sampling and Quantization

- ✓ In practice, the method of **sampling** is determined by the **sensor arrangement** used to generate the image.
  - ❖ **Single Sensor** – Sampling depends on Motion
  - ❖ **Strip Sensor** – Number of Sensor and Motion
  - ❖ **Array Sensor** – Number of Sensor
- ✓ The following figure 4 illustrates **Quantization of the sensor outputs.**

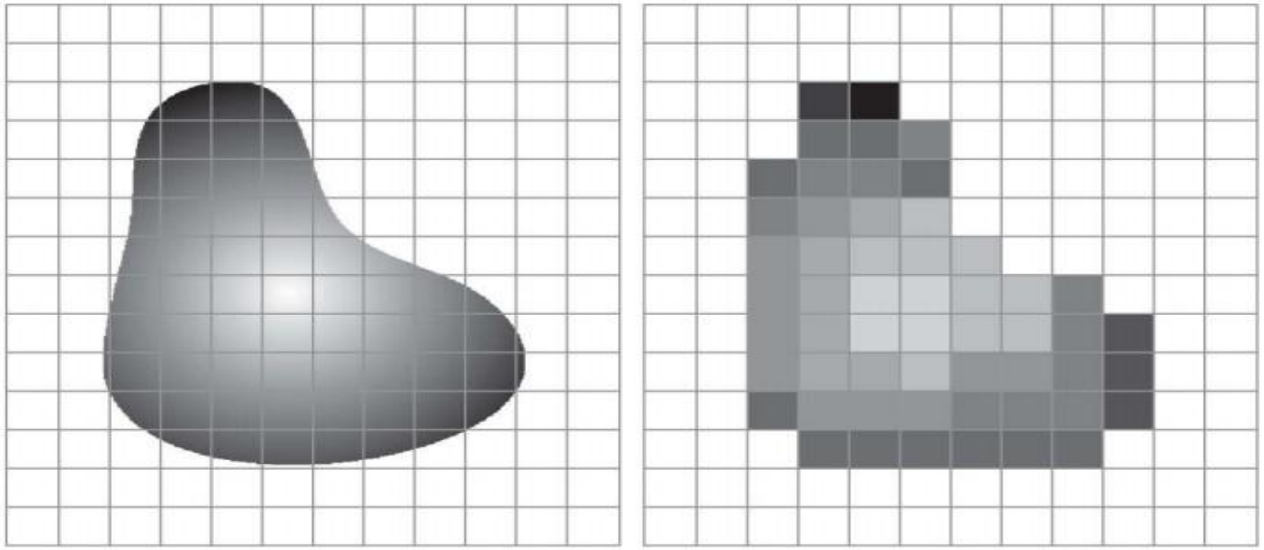
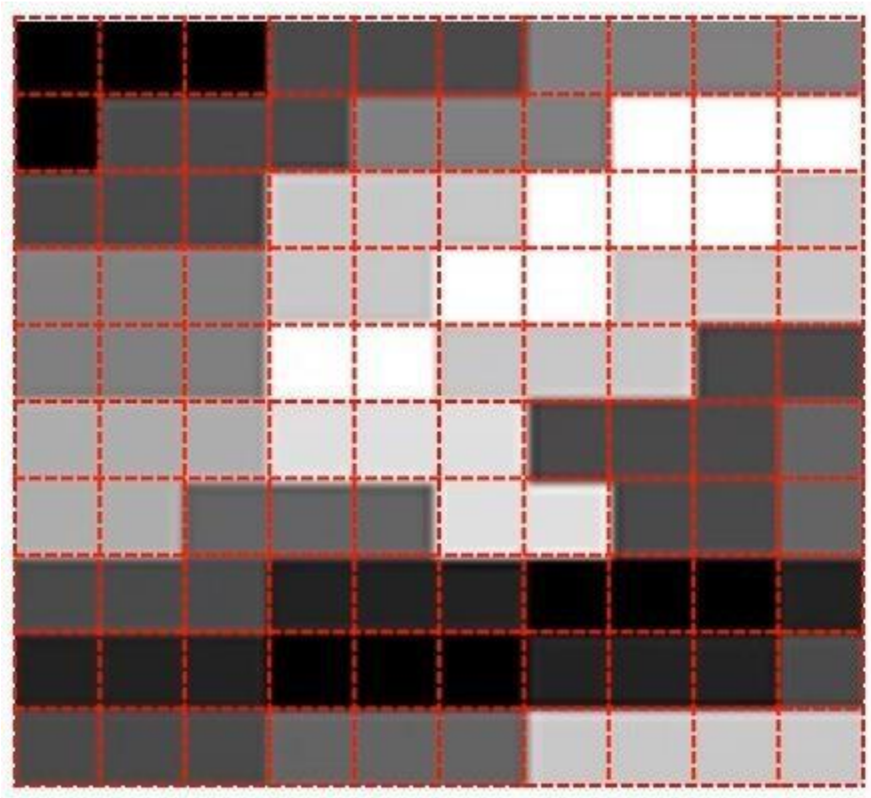


Figure 4: (a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization

Sampling and Quantization



0	0	0	75	75	75	128	128	128	128
0	75	75	75	128	128	128	255	255	255
75	75	75	200	200	200	255	255	255	200
128	128	128	200	200	255	255	200	200	200
128	128	128	255	255	200	200	200	75	75
175	175	175	225	225	225	75	75	75	100
175	175	100	100	100	225	225	75	75	100
75	75	75	35	35	35	0	0	0	35
35	35	35	0	0	0	35	35	35	75
75	75	75	100	100	100	200	200	200	200

Figure 5: Effect of sampling and quantization

Sampling and Quantization

❖ **It determines the spatial resolution of the digital images.**

Sampling and Quantization

- **It determines the number of the gray levels in the image.**

Sampling and Quantization

Clearly, the **quality** of a digital image is determined to a large degree by **the number of samples and discrete intensity levels** used in sampling and quantization.

Representation of Digital Image

f(s,t) - A continuous image function of two continuous variables, **s** and **t.**

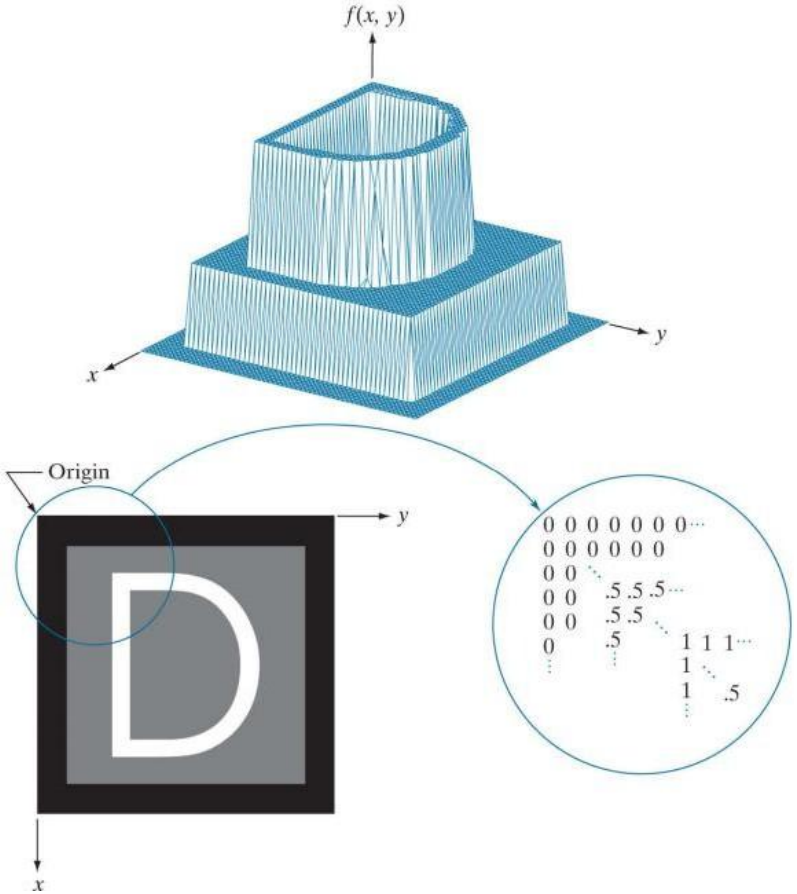
→ Sampling and Quantization →

A 2-D array, *f(x,y)*, containing M rows and N columns, where are discrete coordinates.

- ✓ In general, the value of the image at any coordinates **(x,y)** is denoted **f(x,y),** where x and y are **integers**.
- ✓ The section of the **real plane** spanned by the coordinates of an image is called the **spatial domain**, with **x** and **y** being referred to as **spatial variables** or **spatial coordinates**.

Representation of Digital Image

Figure 6: (a) Image plotted as a surface. (b) Image displayed as a visual intensity array. (c) Image shown as a 2-D numerical array. (The numbers 0, .5, and 1 represent black, gray, and white, respectively.)



Representation of Digital Image

- ✓ This type of representation has three axis. Where, x and y are the spatial co-ordinates and f(x, y) represents another co-ordinates.
- ✓ Here, f(x, y) is a function of spatial co-ordinates (x, y).
- ✓ This type of representation is not always preferable.
- ✓ Complex images having too many details, make the interpretation of images with this type of representation very difficult.

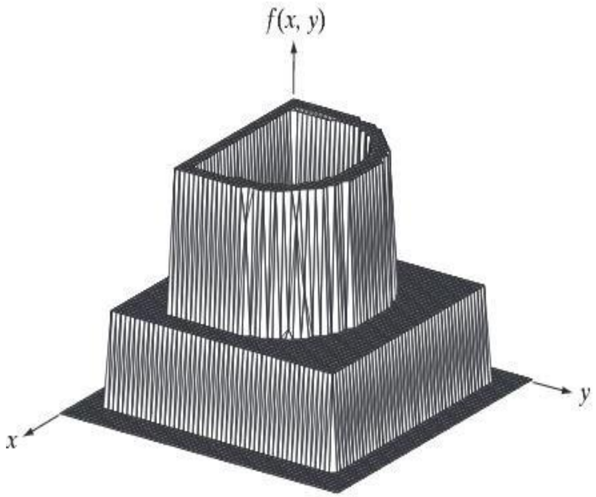


Figure 7: Image plotted as surface

Representation of Digital Image

- ✓ It would appear on a monitor or photograph.
- ✓ There are only **three discrete intensity** values (black, grey and white). If the intensity is **normalized** to the interval **[0, 1]**, then each point in the image has the value ***0, 0.5, or 1***.
- ✓ This type of representation is used for printers.

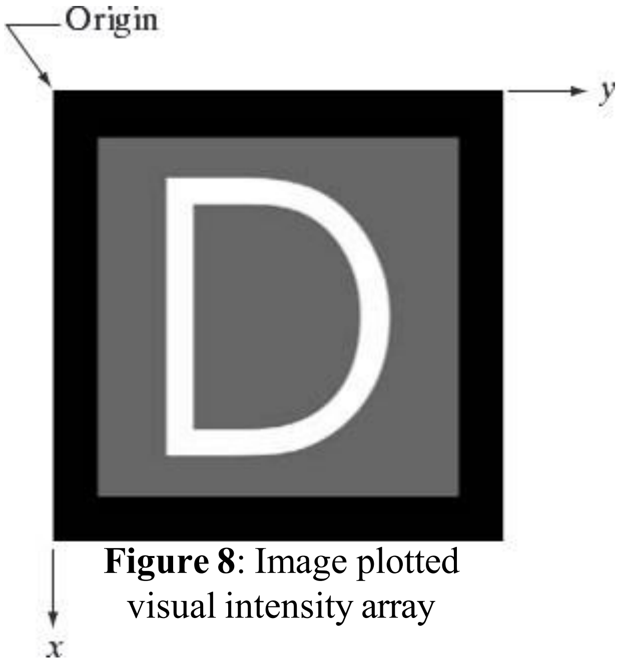


Figure 8: Image plotted visual intensity array

Representation of Digital Image

- ❖ Representation is simply to display the numerical values of as an array (matrix).
- ❖ Most widely used representation in digital image processing.

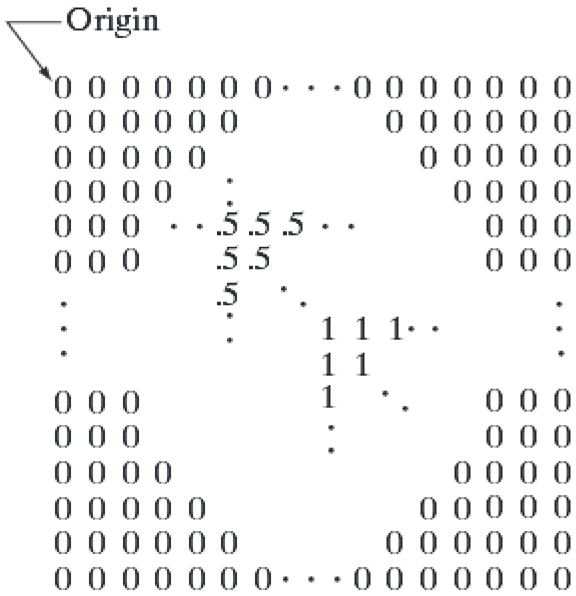


Figure 9: Image shown as a 2-D numerical array (0, 0.5, and 1 represent black, gray, and white, respectively).

Representation of Digital Image

- ✓ Image can also be represented as matrix.
- ✓ Each element of this matrix is called an ***image element, picture element, pixel, or pels***.

Note:
Image – a digital image
Pixel – its elements.

$$f(x,y)=\begin{bmatrix} f(0,0) & f(0,1) & \cdot & f(0,N-1) \\ f(1,0) & f(1,1) & \cdot & f(1,N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & \cdot & f(M-1,N-1) \end{bmatrix}_{M\times N}$$

Representation of Digital Image

- ✓ However, due to **storage and quantizing hardware considerations**, the number of **intensity levels** typically is an integer **power of 2**:

$$L = 2^k$$

- ✓ Interval [0 to L-1].

- ❖ **Dynamic intensity range =** Maximum measurable intensity level / minimum detectable intensity level.
- ❖ **Maximum measurable intensity level** is called ***saturation,*** this means that the intensity level beyond this saturation level is clipped off.
- ❖ ***M*inimum detectable intensity level** is called noise, this means that the intensity level below this noise level is not a true intensity value.

Representation of Digital Image

- Sometimes, the range of values spanned by the gray scale is referred to informally as the **dynamic range**.
- We **define** the **dynamic range of an imaging system** to be the ratio of the maximum measurable intensity to the minimum detectable intensity level in the system.
- As a rule, the upper limit is determined by ***saturation*** and the lower limit by ***noise.***
- Closely associated with this concept is image ***contrast***, which we define as the difference in intensity between the highest and lowest intensity levels in an image.

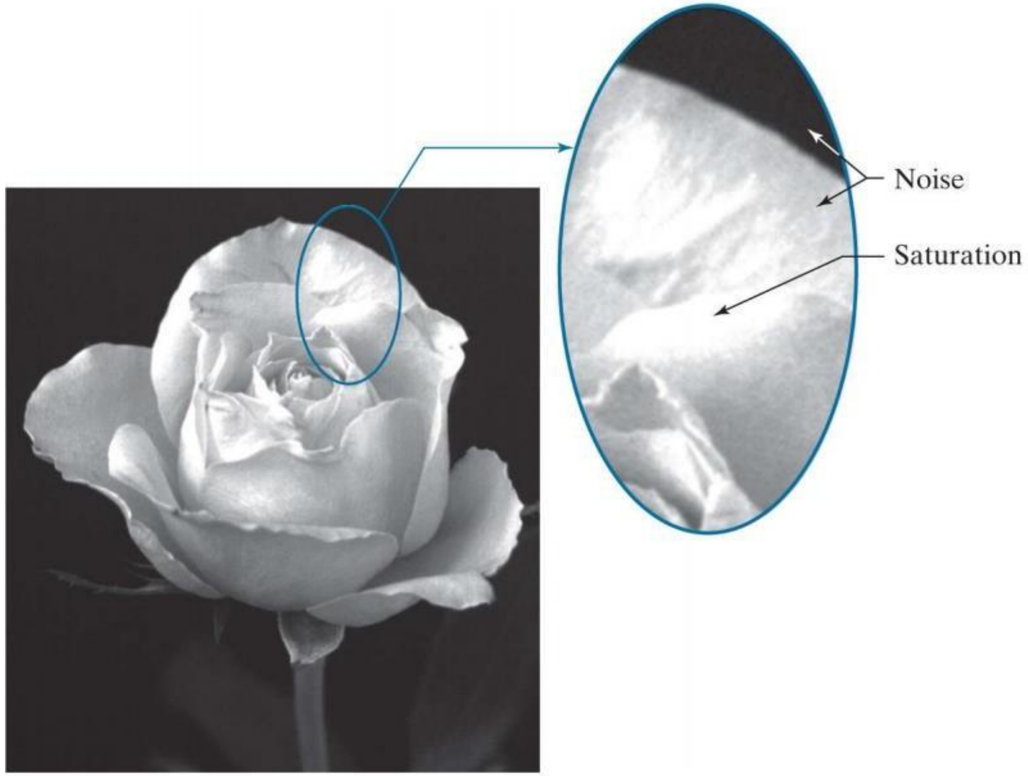


Figure 10: Saturation and Noise

Representation of Digital Image

- When an appreciable number of pixels in an image have a **high dynamic range,** we can expect the image to **have high contrast**.

- Conversely, an image with **low dynamic range** typically has a **dull, washed-out gray look**.

Thank You ☺